

SAN ANDRES, Colombia

WMO: 800010

Lat: 12.584N

Lon: 81.711W

Elev: 19

StdP: 14.69

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	75.0	76.3	67.8	102.1	79.2	69.4	108.0	79.7	22.7	78.6	20.8	79.4	7.9	10	0.445

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.7	88.6	80.7	88.1	80.7	87.7	80.7	82.5	86.3	81.9	85.7	81.4	85.3	9.0	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.0	161.1	86.0	80.7	159.7	85.8	80.3	157.5	85.4	46.2	87.5	45.7	86.6	45.2	85.7	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.6	18.0	16.5	70.1	92.7	2.7	1.9	68.2	94.0	66.6	95.1	65.1	96.2	63.2	97.6		
(5)			66.2	83.5	2.0	0.8	64.7	84.1	63.5	84.5	62.4	84.9	60.9	85.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.4	80.8	81.0	81.4	82.8	83.1	83.1	83.3	83.4	83.3	82.5	82.0	81.6		
	DBStd	1.87	1.46	1.29	1.37	1.43	1.55	1.89	1.57	1.70	1.70	1.90	1.81	1.77		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD50	11813	954	869	974	984	1028	993	1032	1034	998	1006	961	980		
	CDD65	6338	489	449	509	534	563	543	567	569	548	541	511	515		
	CDH74	71441	4885	4470	5204	5975	6618	6443	6812	6879	6520	6235	5759	5641		
	CDH80	21291	837	796	1087	1762	2218	2216	2411	2493	2361	2034	1656	1420		
(14) Wind	WSAvg	8.6	10.3	9.9	8.9	8.2	8.1	9.3	10.7	7.8	5.5	5.7	8.2	10.6		
(15) Precipitation	PrecAvg	74.4	3.2	1.4	1.0	1.0	5.4	7.6	6.9	7.4	9.0	12.5	12.8	6.1		
	PrecMax	107.0	6.9	3.7	3.4	5.0	14.4	24.3	16.8	25.0	16.1	25.1	25.7	15.4		
	PrecMin	46.1	0.5	0.0	0.1	0.0	0.1	1.3	1.6	1.7	5.1	3.2	3.8	1.4		
	PrecStd	15.5	1.7	0.9	0.8	1.1	3.8	4.4	3.5	5.3	3.1	5.5	5.5	3.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.9	86.3	86.4	88.7	88.9	88.8	88.2	89.4	89.8	89.6	88.1	87.0		
		MCWB	77.9	76.9	77.7	80.0	80.5	81.0	80.8	80.8	80.9	80.8	80.1	78.9		
	2%	DB	84.5	85.1	86.0	87.7	87.9	87.9	87.8	88.0	88.2	88.0	87.0	86.2		
		MCWB	77.2	76.6	77.7	79.8	80.5	80.8	80.8	80.9	80.9	80.8	79.7	78.7		
	5%	DB	84.2	84.4	84.7	86.3	86.4	86.4	86.4	87.5	87.8	87.5	86.1	85.6		
		MCWB	77.1	76.7	77.3	78.9	79.9	80.4	80.3	80.8	81.0	80.6	79.3	78.5		
	10%	DB	83.3	83.9	84.3	85.8	86.0	86.0	86.1	86.3	86.9	86.2	85.6	84.4		
		MCWB	76.8	76.6	77.2	78.7	79.8	80.3	80.2	80.5	80.7	80.0	79.1	78.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.4	79.3	80.7	82.2	82.5	82.6	83.1	83.0	83.4	82.7	81.5	81.0		
		MCDB	83.0	83.5	84.5	86.7	86.3	86.4	86.6	86.6	87.2	86.7	85.2	84.3		
	2%	WB	78.9	78.7	79.4	80.8	81.5	81.7	81.7	81.9	82.3	81.7	80.7	80.0		
		MCDB	82.8	83.1	83.6	85.2	85.4	85.5	85.3	85.6	86.2	85.9	84.5	83.6		
	5%	WB	78.3	78.0	78.6	80.0	80.8	81.0	81.0	81.3	81.6	80.9	80.1	79.4		
		MCDB	82.4	82.6	83.1	84.5	84.8	84.7	84.6	85.1	85.6	85.2	84.0	83.4		
	10%	WB	77.6	77.3	77.9	79.5	80.2	80.5	80.5	80.8	80.9	80.3	79.7	79.0		
		MCDB	81.9	82.0	82.7	84.1	84.2	84.2	84.2	84.6	85.1	84.6	83.6	83.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.9	6.1	6.2	6.7	6.2	6.1	6.1	6.7	7.9	8.0	6.6	6.0		
		MCDBR	6.3	6.8	6.7	7.3	6.8	6.5	6.5	7.5	8.6	8.9	7.3	6.7		
	5% WB	MCWBR	3.9	3.9	4.1	4.0	3.9	3.9	4.0	4.4	4.6	5.0	4.3	3.9		
		MCWBR	5.9	5.9	6.1	6.6	6.4	6.1	6.1	6.6	7.8	8.1	6.8	5.9		
(40) Clear-Sky Solar Irradiance	taub	taub	0.393	0.400	0.398	0.448	0.469	0.498	0.508	0.472	0.438	0.395	0.399	0.398		
		taud	2.539	2.506	2.499	2.356	2.330	2.249	2.201	2.337	2.432	2.599	2.571	2.551		
	Ebn at Noon	Ebn at Noon	289	290	292	274	264	254	252	264	276	288	285	284		
		Edh at Noon	32	35	36	41	41	44	46	41	37	31	31	31		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1609	1879	2066	2082	1812	1614	1715	1763	1647	1461	1347	1472		
	RadStd	RadStd	101	61	68	88	143	125	113	133	124	125	175	134		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page